# Claude 3 Model Card

## Methodological Issues

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# Issue 1: Opaque / Under-Justified ASL Thresholds

- **Description**
  - Thresholds (e.g., ASL-3 triggers) are stated but lack explanation or decision analysis.
  - Numeric rules assign high-risk status without explaining why these numbers were chosen.
- **Risks**
  - Policy and safeguard decisions depend on these thresholds.
  - Unclear or unjustified rules can lead to inappropriate or harmful actions.
- **Source**
  - Section 6.2.1 - *Autonomous Replication and Adaptation (ARA) evaluations*

# Issue 2: Generalization of Safety Claims to Models

- **Description**
  - Safety conclusions for entire model family drawn from results on one variant/version.
- **Risks**
  - Non-representative sampling - other variants may differ in safety profile.
  - Version drift - later models can gain harmful capabilities even if earlier ones were “safe.”
  - Generalizing from one variant risks false reassurance.
- **Source**
  - Section 6.2 - Evaluation Results

# Issue 3: Binary Metrics and Poor Reporting

- **Description**
  - Use of pass/fail thresholds without severity levels.
  - Reporting point estimates without confidence intervals or power analysis.
- **Risks**
  - Masks partial progress, variability, and edge cases.
  - No distinction between minor and high-impact harmful outputs.
  - Using current model weaknesses (e.g., hallucinations) as proof of safety is fragile - future models may overcome them.
- **Source**
  - Section 6.2.1 - ARA evaluations, Section 7.1.1 - Multimodal Policy Red-Teaming

# Issue 4: Insufficient Sample Sizes

- **Description**
  - ARA evaluation tasks have very few trials; per-task counts not reported.
  - No clarity on which tasks had reduced sampling, no details on custom environments.
- **Risks**
  - Results may be due to randomness, reducing reliability.
  - Small or missing sample-size data prevents assessment of reproducibility.
- **Source**
  - Section 6.2.1 - ARA evaluations

# Issue 5: Flawed Comparator in Biological Evaluations

- **Description**
  - Google search vs LLM comparison mismatched - retrieval results and links vs synthesised, actionable outputs.
  - Different prompt/context depth and interactivity
  - Human uplift trials introduce sampling noise and operator variation
- **Risks**
  - Small prompt or engineering changes can swing results.
  - Low thresholds not tied to quantified harm or decision rules.
  - Missing per-task counts and stopping rules hinder audits
- **Source**
  - Section 6.2.2 - Biological evaluations

# Issue 6: Missing Human Rater Transparency and Agreement

- **Description**
  - No details on rater selection, inclusivity, training, demographics, or inter-rater agreement.
  - “Security experts” mentioned without specifying numbers, backgrounds, or expertise.
  - No clarity on attempted tasks, task counts, or “custom environments.”
- **Risks**
  - Cannot assess representativeness or reliability of ratings.
- **Source**
  - Section 6.2.3 - Cyber Evaluations, also Section 7.3.1 - Discrimination
- *Model failed safety thresholds but was still released, issues were downplayed*

# Issue 7: Undocumented Classifier Training

- **Description**
  - Classifier for detecting AUP violations lacks details on:
    - Training type (supervised, semi-supervised, etc.)
    - Dataset size, source, composition
    - Annotation process, rater backgrounds, inter-rater agreement
    - Model architecture, hyperparameters, evaluation metrics
- **Risks**
  - Unknown bias, coverage, and failure modes.
  - Poor labeling could miss violations or over-flag benign cases.
- **Source**
  - Section 7 - Policy & Enforcement

# Issue 8: Missing Prompt Sensitivity Analysis

- **Description**
  - No systematic study of how prompt wording, temperature, or framing affect red-teaming results.
- **Risks**
  - Findings may depend on small changes in specific prompts, input order, or framing.
  - Without sensitivity checks, robustness is unknown - findings may depend on specific prompt choices.
- **Source**
  - Section 7.1.1 - Multimodal Policy Red-Teaming

# Issue 9: Gaps in Discrimination Evaluation and Reporting

- **Description**
  - Discrimination evaluation (cited paper) lacks diversity, limited.
  - Plots show some demographic groups favored (score > 0); high scores indicate less equal treatment.
  - Bias plots split into separate comparisons, obscuring full bias progression.
- **Risks**
  - Fairness may not improve consistently; possible regressions in some tiers.
  - Pairwise plots allow selective improvement claims without addressing regressions.
- **Source**
  - Section 7.3 - Societal Impacts (Discrimination)

| Rank | Issue | Justification |
|---|---|---|
| 1 | Binary Metrics & Poor Reporting | Masks severity and uncertainty-critical for safety decision-making |
| 2 | Gaps in Discrimination Evaluation and Reporting | Poses widespread societal harm risks, undermines trust in reporting |
| 3 | Missing Human Rater Transparency and Agreement | Lacks accountability |
| 4 | Opaque / Under-Justified ASL Thresholds | Limits auditable governance |
| 5 | Generalization of Safety Claims to Models | Overlooks variant-specific risks |
| 6 | Missing Prompt Sensitivity Analysis | Reduces robustness to prompt framing |
| 7 | Insufficient Sample Sizes | Undercuts statistical validity and reproducibility |
| 8 | Flawed Comparator in Bio Evaluations | Skews evaluation baselines |
| 9 | Undocumented Classifier Training | Obscures reliability of safety filters |

# Project Proposals

# Proposal 1: Multicultural Discrimination Benchmarking *(to solve Gaps in Discrimination Evaluation)*

## Goals

- Design a multicultural and diverse benchmark with bias probes that cover culturally relevant protected classes and stereotypes.
- Make human rater practice transparent and auditable: publish rater recruitment, training, demographic metadata, and inter-rater reliability statistics.

## Approach

- **Co-design Multilingual Bias Prompts**

  Work with regional partners to design prompts that reflect local sensitive categories (e.g., religion, caste, regional dialects, tribal status, migrant status, LGBTQ+ terms as locally used).

  **Deliverable**: language/culture modules (prompt set + context guide) ready for annotation.

- **Transparent Rater Pipeline with Reliability Metrics**

  Recruit and train raters from target linguistic groups, document their demographics, and measure inter-rater agreement (e.g., Cohen's κ, Krippendorff's α) to ensure consistency.

  **Deliverable:** rater handbook + rater metadata file + IAA report.

# Proposal 1: Multicultural Discrimination Benchmarking *(to solve Gaps in Discrimination Evaluation)*

- **Publish Accessible and Auditable Benchmark Artifacts**
  Release prompts, rater guidelines, demographics, agreement statistics, and scripts for evaluation, enabling transparent and versioned audits by internal and external stakeholders.

## Risks & Mitigations

- Cultural blindspots and misalignment in prompts - Iterative reviews and local expert feedback.
- Low inter-rater agreement for sensitive categories - Use pilot rounds and refine guidelines/training.
- Limited resources - Begin with few languages, use LLM-assisted pre-labeling to prioritize human review for high-variance items

# Proposal 2: Graded Safety Metrics *(to solve Binary Metrics, Poor Reporting)*

## Goals

- Replace pass/fail judgments in multimodal safety tests with a simple severity score (0–4) and confidence intervals (CIs) to capture nuance and uncertainty in harmful or policy-violating responses.
- Scale the review process by combining LLM consensus with targeted human oversight.

## Approach

- **Define a Clear Severity Rubric (0–4):**
  0: Harmless or safe refusal  1: Vague, not helpful  2: Mildly harmful or misleading  3: Clearly harmful, actionable advice  4: Extremely dangerous, detailed, policy-violating content.
- **Annotate using Automated LLMs Labeling with Human Review for Ambiguity**
  Run each prompt-response pair through multiple LLMs, accept consensus when confident, and send cases with disagreement to a small human panel (3 to 5) for final rating.

# Proposal 2: Graded Safety Metrics *(to solve Binary Metrics, Poor Reporting)*

- **Compute Severity Distribution with Confidence Intervals and Publish**
  Use bootstrap resampling to calculate mean severity ± 95% CI across the red-team suite.
  **Deliverable**: Audit bundle with severity rubric, results (with CIs), code for annotation pipeline, and analysis scripts.

## Risks & Mitigations

- LLM consensus may be overconfident - Human reviewers intervene when confidence is low or models disagree.
- Bias in LLM decisions - Using multiple LLMs with different training may help reduce systematic bias.
- Rater variation for complex prompts - A small panel with a clear rubric reduces inconsistency; disagreement flags prompt human review.

# Proposal 3: Transparent ASL Thresholds Framework *(to solve Opaque / Under-Justified ASL Thresholds)*

## Goals

- Replace opaque ASL trigger rules with empirically grounded, risk-based thresholds, ensuring they aren't purely internal or arbitrary.
- Establish a process that is transparent, auditable, and regulator-friendly.

## Approach

- **Develop Quantitative Harm Models**
  Use decision-theoretic methods (e.g., Bayesian decision analysis) to map model capabilities (CBRN or cyber risk potential) to harm probability × severity. **Deliverable:** Capability-to-risk mapping model with harm-curves supporting ASL-3 threshold placement.
- **Integrate Regulatory Mapping & Principles**
  Cross-reference ASLs with definitions of "high-risk AI" under the AI Act, and analogous concepts in UK/US frameworks. **Deliverable:** Flexible mapping document showing how ASL labels align with external risk classifications.

# Proposal 3: Transparent ASL Thresholds Framework *(to solve Opaque / Under-Justified ASL Thresholds)*

- **Design Governance & Transparency Protocols**
  Publish detailed rationale, sensitivity analyses, harm-curve visualizations, and expert elicitation process. **Deliverable**: A governance design template for threshold decision-making.

**Risks & Mitigations**

- Catastrophic AI harms are hard to quantify - Mitigate with conservative buffers, expert diversity, and iterative refinement.
- Jurisdictions differ in legal frameworks - Use flexible mapping rather than rigid equivalences, and emphasize adaptability.
- Labs may resist external constraints - Mitigate by co-designing thresholds collaboratively and highlighting alignment benefits for deployment clarity.

# Other Issues in Model Card [1/2]

- **Circular or Brittle Grading Methods**
  - Safety judgments sometimes rely on automated classifiers or brittle string-matching for refusals.
  - Risk: Creates evaluator bias, misses nuanced harmful outputs.
  - Ref: Sec 7.1 & 5.4.1 – Trust & Safety methodology, refusal metrics.

- **Reliance on Internal / Non-public Benchmarks**
  - Key evaluations (e.g., "100Q Hard," internal agentic coding) are not publicly released.
  - Risk: Limits reproducibility and independent validation.
  - Ref: Sec 3.2 – Internal evaluation references.

# Other Issues in Model Card [2/2]

- **Metrics Without Statistical Uncertainty**
  - Percentages and recalls reported as single values without confidence intervals or sample sizes.
  - Risk: Decision-makers cannot assess stability or significance.
  - Ref: Sec 2 – Result tables and metrics.

- **Guardrail (HHH) Training May Under-Elicit Capabilities**
  - Safety tuning may cause the model to decline responses, hiding raw capabilities.
  - Risk: Underestimates potential misuse or hidden capabilities.
  - Ref: Sec 3.2 – Safety Evaluations Overview.